Markov Decision Processes

2026-06-07

1 : MDP

MDP **(fully observable)** . " " Markov , **(reward)** **(action)** .

$$\underbrace{\langle S, P\rangle}_{\text{Markov Process}} \xrightarrow{+R,\,\gamma} \underbrace{\langle S, P, R, \gamma\rangle}_{\text{MRP}} \xrightarrow{+A} \underbrace{\langle S, A, P, R, \gamma\rangle}_{\text{MDP}}$$

> **!**
>
> **MDP** π **MRP** **, MRP** **Markov Process** **.** (evaluation) " MRP " .

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2 Markov Process

2.1 Markov

$$\mathbb{P}[S_{t+1} \mid S_t] = \mathbb{P}[S_{t+1} \mid S_1, \ldots, S_t]$$

S_t **(sufficient statistic)** . history .

2.2

$$P_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s], \qquad \sum_{s'} P_{ss'} = 1 \;(\quad = 1)$$

Markov Process (Chain) $\langle S, P\rangle$ — S P . .

2.3 Student Markov Chain

(sample episode) : `C1 → C2 → C3 → Pass → Sleep`, `C1 → FB → FB → C1 → C2 → Sleep` .

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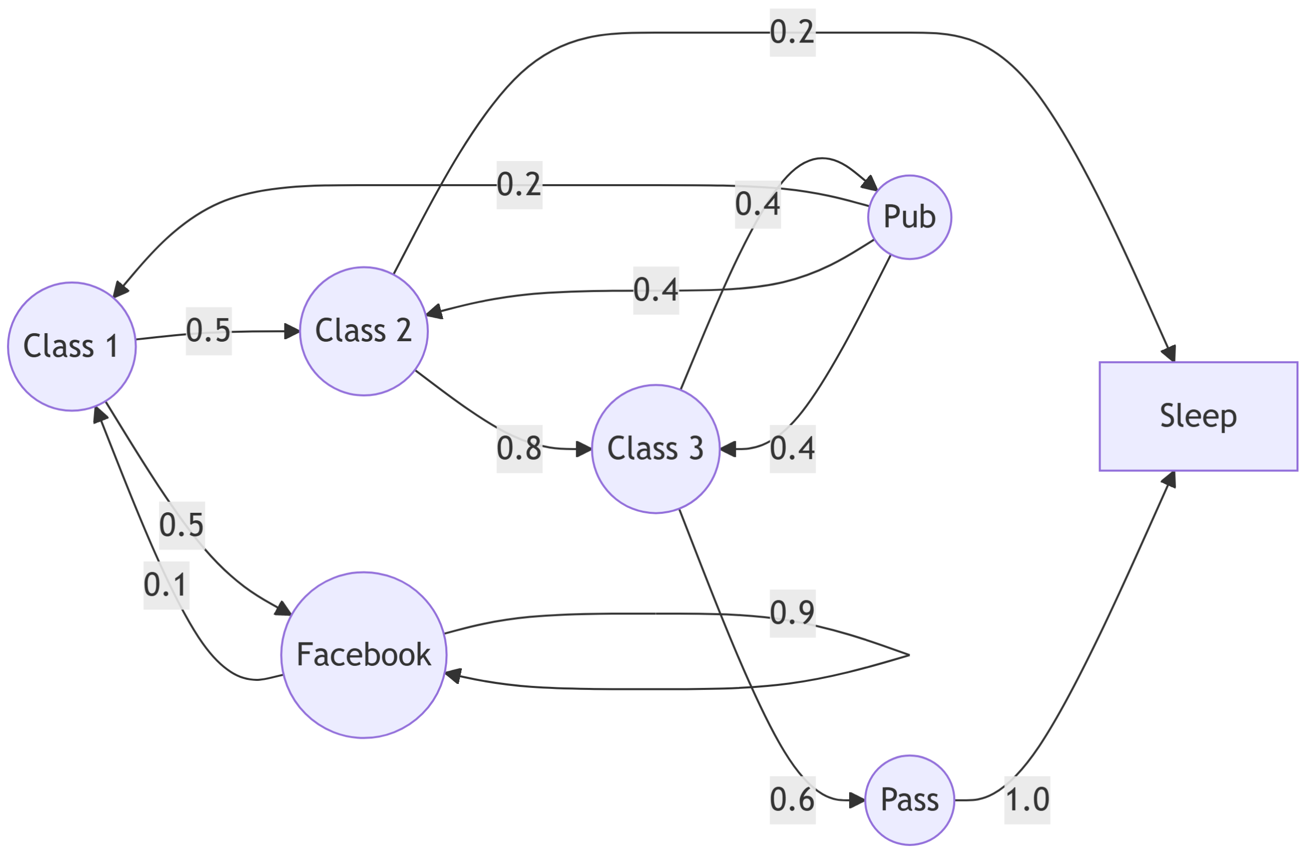
3 Markov Reward Process (MRP)

MRP $= \langle S, P, R, \gamma\rangle$ — Markov chain **(value)** .

- : $R_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
- : $\gamma \in [0, 1]$



1: Student Markov Chain

3.1 Return ( )

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

> **i**
>
> s R_{t+1} . G_t , $v(s)$.

3.2 — γ

- ( )
- (cyclic) Markov process **return**
-
-
- /
- (terminate) $\gamma = 1$ (undiscounted)

3.3 Bellman Equation

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

Return + .

$$\begin{aligned} v(s) &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma\, v(S_{t+1}) \mid S_t = s] \\ &= R_s + \gamma \sum_{s'} P_{ss'}\, v(s') \end{aligned}$$

$\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = v(s')$ tower property .

3.4

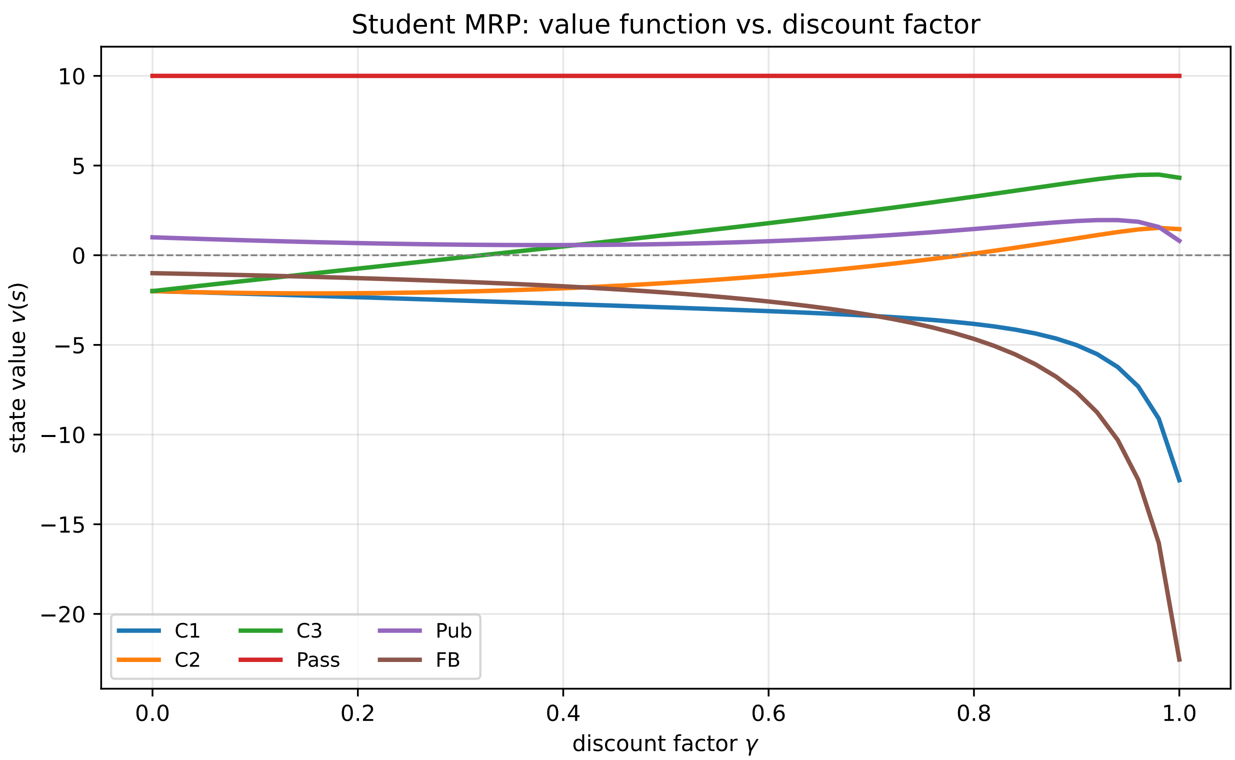
$$v = R + \gamma P v \implies (I - \gamma P)\, v = R \implies v = (I - \gamma P)^{-1} R$$

- , $O(n^3)$ $(n = |S|)$.
- $\gamma < 1$ $\rho(\gamma P) < 1$ $(I - \gamma P)$. n DP / Monte-Carlo / TD .

3.5 γ

Sleep (terminal, $v = 0$) $\{C1, C2, C3, \text{Pass}, \text{Pub}, FB\}$ $v_T = (I - \gamma Q)^{-1} R_T$. Q , $\gamma = 1$ $\rho(Q) < 1$.

```
import numpy as np
import matplotlib.pyplot as plt

states = ["C1", "C2", "C3", "Pass", "Pub", "FB"]
idx = {s: i for i, s in enumerate(states)}

Q = np.zeros((6, 6))
Q[idx["C1"], idx["FB"]] = 0.5;  Q[idx["C1"], idx["C2"]] = 0.5
Q[idx["C2"], idx["C3"]] = 0.8                    # C2 -> Sleep 0.2 (terminal)
Q[idx["C3"], idx["Pub"]] = 0.4; Q[idx["C3"], idx["Pass"]] = 0.6
# Pass -> Sleep 1.0 (terminal):    0
Q[idx["Pub"], idx["C1"]] = 0.2
Q[idx["Pub"], idx["C2"]] = 0.4
Q[idx["Pub"], idx["C3"]] = 0.4
Q[idx["FB"], idx["FB"]] = 0.9;  Q[idx["FB"], idx["C1"]] = 0.1

R = np.array([-2, -2, -2, 10, 1, -1], dtype=float)

gammas = np.linspace(0.0, 1.0, 51)
V = np.array([np.linalg.solve(np.eye(6) - g * Q, R) for g in gammas])

fig, ax = plt.subplots(figsize=(8, 5))
for j, s in enumerate(states):
    ax.plot(gammas, V[:, j], marker="", linewidth=2, label=s)
ax.axhline(0, color="gray", linewidth=0.8, linestyle="--")
ax.set_xlabel(r"discount factor $\gamma$")
ax.set_ylabel(r"state value $v(s)$")
ax.set_title("Student MRP: value function vs. discount factor")
ax.legend(ncol=3, fontsize=9)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



2: gamma v(s)

$\gamma = 0$ $v(s) = R_s$ ( ) . γ , $R = +10$ Pass $C3, C2$. 0.9 FB γ $R = -1$ ($\gamma = 1$ $v(FB) \approx -22.5$).

γ	C1	C2	C3	Pass	Pub	FB
0.0	-2.00	-2.00	-2.00	$+10.00$	$+1.00$	-1.00
0.5	-2.91	-1.55	$+1.12$	$+10.00$	$+0.62$	-2.08
0.9	-5.01	$+0.94$	$+4.09$	$+10.00$	$+1.91$	-7.64
1.0	-12.54	$+1.46$	$+4.32$	$+10.00$	$+0.80$	-22.54

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4 Markov Decision Process (MDP)

MDP $= \langle S, A, P, R, \gamma \rangle$ — MRP **(decision)** .

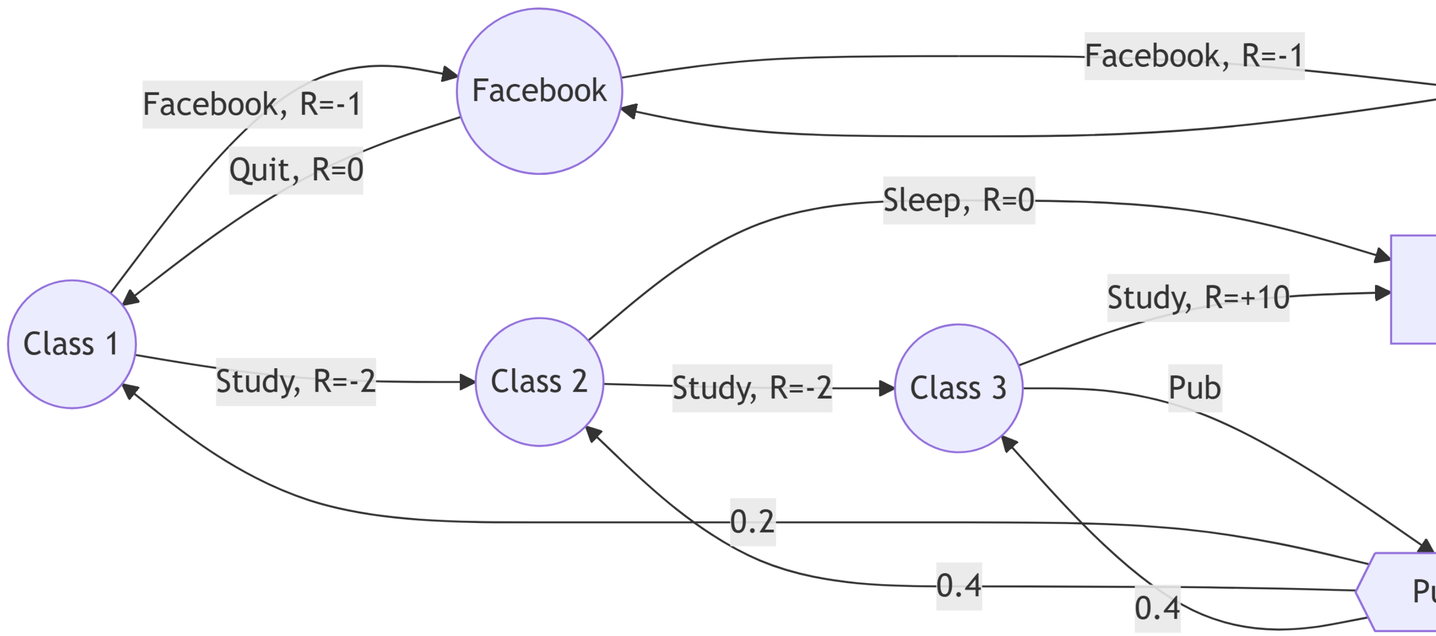
$$P^a_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s,\ A_t = a], \qquad R^a_s = \mathbb{E}[R_{t+1} \mid S_t = s,\ A_t = a]$$

4.1 (policy)

$$\pi(a \mid s) = \mathbb{P}[A_t = a \mid S_t = s]$$

(stationary), . history Markov .

4.2 Student MDP



3: Student MDP (= action / reward)

MRP " " , MDP " " . Pub (0.2/0.4/0.4) , .

4.3 → MRP

π $S_1, S_2, \ldots$ Markov process $\langle S, P^\pi \rangle$, - MRP $\langle S, P^\pi, R^\pi, \gamma \rangle$.

$$P^\pi_{ss'} = \sum_a \pi(a \mid s)\, P^a_{ss'}, \qquad R^\pi_s = \sum_a \pi(a \mid s)\, R^a_s$$

4.4

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s,\ A_t = a]$$

v_π , q_π (,) . .

4.5 Bellman Expectation Equation

v q .

$$v_\pi(s) = \sum_a \pi(a \mid s)\, q_\pi(s,a)$$
$$q_\pi(s,a) = R_s^a + \gamma \sum_{s'} P_{ss'}^a\, v_\pi(s')$$
$$v_\pi(s) = \sum_a \pi(a \mid s)\Big(R_s^a + \gamma \sum_{s'} P_{ss'}^a\, v_\pi(s')\Big)$$
$$q_\pi(s,a) = R_s^a + \gamma \sum_{s'} P_{ss'}^a \sum_{a'} \pi(a' \mid s')\, q_\pi(s',a')$$

:

$$v_\pi = R^\pi + \gamma P^\pi v_\pi \quad \Longrightarrow \quad v_\pi = (I - \gamma P^\pi)^{-1} R^\pi$$

MRP .

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5 (Optimality)

5.1

$$v_*(s) = \max_\pi v_\pi(s), \qquad q_*(s,a) = \max_\pi q_\pi(s,a)$$

MDP " " v_* (q_*) .

5.2

MDP .

1. π_* .
2. $v_{\pi_*}(s) = v_*(s)$.
3. $q_{\pi_*}(s,a) = q_*(s,a)$.

5.3

q_* greedy (deterministic) .

$$\pi_*(a \mid s) = \begin{cases} 1 & \text{if } a = \arg\max_{a \in A} q_*(s,a) \\ 0 & \text{otherwise} \end{cases}$$

!

. q_* MDP .

5.4 Bellman Optimality Equation

$$
\begin{aligned}
v_*(s) &= \max_a q_*(s,a) \\
q_*(s,a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \, v_*(s') \\
v_*(s) &= \max_a \left(R_s^a + \gamma \sum_{s'} P_{ss'}^a \, v_*(s') \right) \\
q_*(s,a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \max_{a'} q_*(s',a')
\end{aligned}
$$

5.5 Expectation vs. Optimality

Bellman **Expectation**	$\sum_a \pi(a \mid s)\,(\cdot)$ ( )	$(I-\gamma P^\pi)^{-1}R^\pi$
Bellman **Optimality**	$\max_a(\cdot)$	$\rightarrow$

max , **Value Iteration, Policy Iteration, Q-learning, Sarsa** .

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6 Extensions to MDPs

6.1 · MDP

- (countably infinite) / :
- / : -2 (LQR)
- : $\rightarrow$ **HJB(Hamilton-Jacobi-Bellman)** , Bellman time-step $\rightarrow 0$

6.2 POMDP ( )

$\langle S, A, O, P, R, Z, \gamma \rangle$. MDP = HMM.

- **Belief state** $b(h)$: history h
- POMDP ( ) **history tree** ( ) **belief state tree**

6.3 Undiscounted / MDP

- **Ergodic** Markov process = recurrent( ) + aperiodic( )
- π chain ergodic MDP ergodic
- ergodic MDP ρ^π
- **average-reward Bellman equation** (differential value function)

7

1. $v = (I - \gamma P)^{-1} R$ — Bellman . (MRP/leaky bucket ·)
2. **Expectation vs. Optimality** **4** — $v \leftrightarrow q$. max vs. $\sum_a \pi$ / .
3. γ **6** .
4. $+ q_*$ greedy .
5. $\rightarrow$ **MRP** $(P^\pi, R^\pi$) — .

leaky bucket · contention "Markov chain " , .